

E19B.1(a) The volume of oxygen gas at 0 °C and 104 kPa adsorbed on the surface of 1.00 g of a sample of silica at 0 °C was 0.286 cm³ at 145.4 Torr and 1.443 cm³ at 760 Torr. Assume that the Langmuir isotherm applies estimate the value of V_{∞} .

E19B.3(a) A certain solid sample adsorbs 0.44 mg of CO when the pressure of the gas is 26.0 kPa and the temperature is 300 K. The mass of gas adsorbed when the pressure is 3.0 kPa and the temperature is 300 K is 0.19 mg. The Langmuir isotherm is known to describe the adsorption. Estimate the fractional coverage of the surface at the two pressures.

E19B.4(a) The adsorption of a gas is described by the Langmuir isotherm with $\alpha = 0.75 \text{ kPa}^{-1}$ at 25 °C. Calculate the pressure at which the fractional surface coverage is (i) 0.15, (ii) 0.95.

E19B.2(a) The enthalpy of adsorption of CO on a surface is found to be -120 kJ mol^{-1} . Estimate the mean lifetime of a CO molecule on the surface at 400 K; take $\tau_0 = 1.0 \times 10^{-14} \text{ s}$.

E19B.5(a) A solid in contact with a gas at 12 kPa and 25 °C adsorbs 2.5 mg of the gas and obeys the Langmuir isotherm. The enthalpy change when 1.00 mmol of the adsorbed gas molecules is desorbed is +10.2 J. What is the equilibrium pressure for the adsorption of 2.5 mg of gas at 40 °C? *Hint:* The data are isosteric; use a similar approach to that in *Example 19B.2*.

E19B.8(a) The average time for which an oxygen atom remains adsorbed to a tungsten surface is 0.36 s at 2548 K and 3.49 s at 2362 K. What is the activation energy for desorption?

25.4b A monolayer of CO molecules (effective area 0.165 nm^2) is adsorbed on the surface of 1.00 g of an $\text{Fe/Al}_2\text{O}_3$ catalyst at 77 K, the boiling point of liquid nitrogen. Upon warming, the carbon monoxide occupies 4.25 cm^3 at 0°C and 1.00 bar. What is the surface area of the catalyst?